

25 mm round tag. Coil on the front copper; electronics inside the coil; wired-interface pads on the back.

The master device ID (hwid:...), and its Data Matrix code are marked on the exposed face (PBS-HW-ID-01 REQ-031).

Unpowered most of its life; the contactless front end powers the secure element from the reader's field.

Secure element (Ed25519, on-chip key generation, I2C)

ISO/IEC 15693 contactless front end with energy harvest

Current limiter (latch-up protection)

Power OR-ing (dual ideal diode)

Antenna tuning capacitor

Contactless antenna coil (copper spiral)

Interface pads: 3.3 V, GND, SCL, SDA (PBS-HW-ID-01 Section 5.3)

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PBS-HW-ID-01 open reference design; generic placeholders, no part numbers
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Sheet: /
File: pbs-hwid-tag.kicad_sch
Title: PBS-HW-ID-01 master device identity tag: Identity tag
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U1

V_SE 1 VDD LINK 5 SE_LINK

SCL 2

SDA 3

GND 4

Secure element (Ed25519, on-chip key generation, I2C)

U2

COIL_A 1 COIL_A LINK 4 SE_LINK

COIL_B 2 COIL_B VHARV 5 V_HARV

GND 3

ISO/IEC 15693 contactless front end with energy harvest

U3

V_IN 1 IN OUT 3 V_PROT

GND 2

Current limiter (latch-up protection)

U4

V_PROT 1 A OUT 3 V_SE

V_HARV 2 B GND 4 GND

Power OR-ing (dual ideal diode)

C1

COIL_A 1 A B 2 COIL_B

Antenna tuning capacitor

L1

COIL_A 1 A B 2 COIL_B

Contactless antenna coil (copper spiral)

J1

V_IN 1 3V3 SCL 3 SCL

GND 2 GND SDA 4 SDA

Interface pads: 3.3 V, GND, SCL, SDA (PBS-HW-ID-01 Section 5.3)

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